

Agenda

1. Announcements

- 9.2, 9.3, 9.4 HW due today
- Discussion Quiz 5 tomorrow on
- Exam Solutions posted on Canvas
- Exam grades will be released tomorrow

2. Section 11.1

3. Section 11.2

Useful Theorems

Theorem Let $\{a_n\}$ be a geometric sequence where $a_n = a_1 r^{n-1}$ for constants a_1 and r

1.) If $-1 < r < 1$ then $\lim_{n \rightarrow \infty} a_n = 0$

2.) $r = 1$, then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_1 = a_1$

3.) $r = -1$, then $\lim_{n \rightarrow \infty} a_n$ DNE

4.) $r > 1$, then $\lim_{n \rightarrow \infty} a_n = \pm \infty$ 5. $r < -1$ DNE

Activity

1.) $\lim_{n \rightarrow \infty} 5\left(\frac{2}{3}\right)^{n-1} = 0$
 $r = 2/3$
converges

2.) $\lim_{n \rightarrow \infty} 15 = 15$
 $r = 1$
converges

3.) $\lim_{n \rightarrow \infty} (-2)^n = \text{DNE}$
 $r = -2$
diverges

Theorem: If $\lim_{n \rightarrow \infty} |a_n| = 0$, then $\lim_{n \rightarrow \infty} a_n = 0$

Example: $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n}$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \text{so} \quad \lim_{n \rightarrow \infty} \frac{-1^n}{n} = 0$$

Monotonic Sequence Theorem

normal in.
 $a_1 \leq a_2 \leq a_3 \leq \dots$

Definitions: Let $\{a_n\}$ be a sequence

$a_n < a_{n+1}$ for all n

1.) It is called monotonically increasing if $a_1 < a_2 < a_3 < \dots$

2.) It is called monotonically decreasing if $a_n > a_{n+1}$ for all n

3.) It is called bounded if there is a constant M

such that $|a_n| \leq M$ for all $n \geq 1$

Example:

1.) $a_n = -2 + 3n$

increasing
not bounded

diverge

2.) $b_n = \cos(n)$

not monotonic
bounded $M = 1$

diverge

3.) $a_n = 1 + \frac{1}{n} = \frac{n+1}{n}$

decreasing
bounded $M = 2$

$$1 < a_n \leq 2$$

converge

Monotonic Sequence Theorem:

Every bounded monotonic sequence converges

Example: Let $\{a_n\}$ be the sequence defined by

$$a_1 = 1, \quad a_{n+1} = 4 - \frac{2}{a_n}$$

1.) Write out the first few terms of the sequence.

$$a_1 = 1, \quad a_2 = 4 - \frac{2}{a_1} = 4 - \frac{2}{1} = 2, \quad a_3 = 4 - \frac{2}{a_2} = 4 - \frac{2}{2} = 3$$

$a_n = \frac{1}{2}$

2.) Assume that $\{a_n\}$ is monotonically increasing and bounded.

What is $\lim_{n \rightarrow \infty} a_n$?

$$L = \lim_{n \rightarrow \infty} a_n \quad \text{therefore} \quad L = \lim_{n \rightarrow \infty} a_{n+1}$$

$$L = \lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \left(4 - \frac{2}{a_n} \right) = 4 - \frac{2}{\lim_{n \rightarrow \infty} a_n} = 4 - \frac{2}{L}$$

$$L = 4 - \frac{2}{L}$$

$$L^2 = 4L - 2$$

$$L^2 - 4L + 2 = 0$$

$$L = \frac{4 \pm \sqrt{16 - 8}}{2}$$

$$L = 2 \pm \sqrt{2}$$

$$L \neq 2 - \sqrt{2} > 1$$

$$2 - \sqrt{2} < 1$$

Series

We will now investigate summing terms together as an infinite series:

Define the n^{th} partial sum as

Activity: Write each infinite series using summation notation

1.) $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$

2.) $\frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \frac{2}{81} + \dots$

3.) $0 + 1 + 0 + 1 + \dots$

Definitions Let $S_n = a_1 + a_2 + a_3 + \dots + a_n$ be the n^{th} partial sum

1.) We say $\sum_{n=1}^{\infty} a_n$ converges if

2.) We say $\sum_{n=1}^{\infty} a_n$ diverges if

Examples: Determine whether each series converges or diverges

1.) $\sum_{n=1}^{\infty} (-1)^n$

2.) $\sum_{n=1}^{\infty} 0$

3.) $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$